Data Analysing Report

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1. Introduction

1.1 Context

The use of Artificial Intelligence systems in the financial sector has expanded rapidly across applications such as credit scoring, fraud detection, algorithmic trading, risk assessment, customer service automation, and compliance support. This expansion increases operational capacity, but it also introduces new risk vectors in highly regulated environments where automated decisions can generate material social, legal, and operational consequences.[1]

In financial services, AI failures affect more than technical performance. They may generate reputational damage, discriminatory bias, operational losses, regulatory scrutiny, and internal policy changes, which makes the study of real-world incidents especially relevant for governance and risk management.[1]

This report is based on real incidents documented in the AI Incident Database (AIID) and filtered for the context of financial services. The analytical goal is to transform this source into a structured evidence base capable of supporting statistical analysis, predictive modelling, relational storage, API exposure, and dashboard consumption.[1]

1.2 Research Problem

Given a set of AI incidents recorded across multiple sectors and filtered to the financial domain, the central analytical problem is to assess whether systematic patterns can be observed in the concentration of incidents by application type, in the unequal effect of algorithmic bias across customer segments, in the relationship between severity and regulatory response, and in the usefulness of derived attributes for predictive modelling.[1]

1.3 Practical Relevance

Stakeholder	Direct benefit
Banks and fintechs	Better understanding of operational and reputational risk.[1]
Compliance teams	Support for prioritising audits, reviews, and controls.[1]
Risk managers	Identification of exposure patterns by application type.[1]
Regulators	Evidence for incident-based supervision.[1]
Product and AI teams	Better understanding of the risk factors associated with AI deployment.[1]

2. Research Objectives and Questions

2.1 General Objective

The general objective is to evaluate whether structured data on AI incidents in financial services reveal relevant patterns of algorithmic bias, operational risk, and governance that can support analysis, monitoring, and decision-making.[1]

2.2 Specific Objectives

1. Identify AI incidents related to financial services.[1]
2. Structure and enrich the data with analytically relevant derived variables.[1]
3. Evaluate statistical hypotheses involving concentration, bias, severity, and regulatory response.[1]
4. Build predictive models for severity classification and regulatory investigation prediction.[1]
5. Organise the results into an integrated architecture composed of notebooks, relational storage, a RESTful API, and an interactive dashboard.[1]

2.3 Research Hypotheses

Hypothesis	Research question	Analytical approach
H1	Are incidents concentrated in specific application types?	Chi-square goodness-of-fit test.[1]
H2	Does algorithmic bias affect some customer segments disproportionately?	Chi-square test of independence.[1]
H3	Do more severe incidents trigger stronger regulatory response?	Chi-square analysis and logistic regression.[1]
H4	Is there a temporal trend in incident volume?	Temporal correlation and trend analysis.[1]

3. Data Background and Source Context

3.1 Main Data Source

The main source used in the project is the AI Incident Database (AIID), a repository of real incidents involving AI systems documented from public sources. This source is particularly useful for exploratory studies, risk analysis, and the documentation of socio-technical failures.[1]

The project references three access paths to the source: the official portal, the official GraphQL API, and a structured Kaggle dataset alternative.[1]

3.2 Financial Services Scope

Not every AIID incident belongs to the financial domain, so the project applies thematic filtering based on keywords and textual context. The filter targets incidents related to credit, fraud, banking, fintech, financial markets, investments, insurance, payments, risk management, and algorithmic trading.[1]

3.3 Source Limitations

The source has important limitations because it only records incidents that became public, may reflect media and geographic bias, does not always include complete

financial values, depends heavily on textual richness, and mixes cases of different natures that require filtering and attribute engineering. These limitations do not invalidate the project, but they require careful interpretation of the results and moderation in the conclusions.[1]

4. Analytical Methodology (CRISP-DM)

The project is organised according to the CRISP-DM methodology because it allows the analytical workflow to be structured into clear and cumulative stages.[1]

4.1 Applied Stages

1. Business Understanding: definition of the problem, objectives, financial scope, and research hypotheses.[1]
2. Data Understanding: initial exploration of the source, identification of available columns, reading of textual fields, and assessment of analytical potential.[1]
3. Data Preparation: filtering of the financial subset, standardisation of columns, treatment of data, and creation of derived variables.[1]
4. Modeling: training of supervised models for severity and regulatory investigation.[1]
5. Evaluation: statistical tests, inferential interpretation, and discussion of model quality.[1]
6. Deployment: structuring of the SQLite database, creation of a RESTful API, and conception of an interactive dashboard.[1]

4.2 Pipeline Summary

The workflow presented in the source report moves from AIID, Kaggle, and local cache into Notebook 1 for exploration and preparation, then branches into a treated CSV and

SQLite storage; from the treated CSV, Notebook 2 performs statistical analysis and Notebook 3 performs machine learning, while the serialised models and SQLite database are integrated in Notebook 4 for API serving and dashboard consumption.[1]

5. Data Used and Preparation

5.1 Analytical Base

The processed analytical base used in the subsequent analyses contains 31 incidents from the financial sector and covers the period from 2003 to 2023. This subset is treated as the main dataset for both statistical analysis and modelling.[1]

5.2 Main Variables

The analytical structure combines original fields and derived attributes. The most relevant variables used across the project include incident_id, title, text, year, application_type, incident_type, customer_segment, severity_level, regulatory_investigation, fine_imposed, policy_change, and third_party_audit.[1]

5.3 Attribute Engineering

Data preparation includes column name standardisation, year extraction from dates, consolidation and cleaning of textual fields, duplicate removal, thematic filtering for the financial sector, creation of derived categorical variables, and creation of governance and regulatory flags.[1]

5.4 Why the Treated CSV Was Used

Although the official AIID API is useful for acquisition and updating, the analytical stage depends strongly on the semantic quality of textual fields. Because several derived variables were inferred from title, summary, and text, the project adopts the treated and enriched CSV as the main source for later stages due to its greater consistency and reproducibility.[1]

6. Analytical Variables and Research Hypotheses

6.1 Core Derived Variables

The derived variables were built through heuristic inference over free text, mainly using the title and description/summary fields, which were consolidated into a single lowercase text field to facilitate semantic pattern matching. In Notebook 1, the pipeline renames columns, converts dates, extracts year, and creates the text field by concatenating title and summary after normalisation

The eight main derived variables are application_type, incident_type, customer_segment, severity_level, regulatory_investigation, fine_imposed, policy_change, and third_party_audit.

6.2 Semantic Matching Logic

The report explains that semantic correlation was not implemented through embeddings, TF-IDF, or advanced NLP models. Instead, it was implemented as rule-based semantic matching through keywords and word stems, in which each incident text was compared with lists of representative terms for each class.[1]

Examples include keywords for credit scoring, lending, mortgages, fraud, anti-money laundering, trading, flash crash, risk assessment, bias, discrimination, outages, and data breach scenarios.[1]

6.3 Rule-Based Inference

The inference logic is deterministic and based on if/elif rules, so each incident receives the class associated with the first compatible keyword group found in the text. When textual evidence is insufficient, fallback categories such as otherfinance, other, and general are used.[1]

6.4 Governance Flags

The four governance flags are extracted from the consolidated text using semantic lists related to investigation, fines, policy change, and third-party audit. This procedure transforms unstructured narrative text into structured analytical variables that can later be used in statistics and machine learning.[1]

6.5 Methodological Limitations of Derived Variables

This inference strategy depends strongly on the semantic density of the textual fields, so sparse or incomplete titles, summaries, or descriptions weaken the quality of the derived attributes. The source therefore positions the approach as an interpretable and reproducible heuristic rather than deep semantic understanding, and recommends future evolution toward stronger NLP techniques such as embeddings and large-scale TF-IDF approaches

7. Statistical Analysis and Inferential Findings

Notebook 2 corresponds to the Evaluation phase of CRISP-DM and focuses on descriptive statistics, bivariate analysis, and hypothesis testing around four central hypotheses

7.1 Descriptive Overview

The analysed set contains 31 incidents covering 2003 to 2023. This size is sufficient for exploration and limited inference, but it imposes important constraints on statistical power and the stability of stronger conclusions

7.2 Hypothesis H1

The first hypothesis investigates whether incidents are uniformly distributed across different financial AI application types. From a managerial perspective, this helps identify where governance should be more intensive, such as credit, fraud, or automated trading

7.3 Hypothesis H2

The second hypothesis evaluates whether incidents associated with algorithmic bias affect some customer segments disproportionately. This is central to discussions of algorithmic fairness, indirect discrimination, and asymmetry of impact in financial services

7.4 Hypothesis H3

The third hypothesis tests whether more severe incidents tend to be more associated with regulatory investigation. The notebook also indicates logistic regression as a complementary analytical approach for studying the relationship between incident attributes and regulatory response

7.5 Hypothesis H4

The fourth hypothesis examines whether there is a temporal trend in incident volume. The report notes that this question should be interpreted with caution because the total number of cases is small and the time series is short for robust temporal patterns.[1]

7.6 Critical Interpretation

The main contribution of Notebook 2 is not merely to prove hypotheses, but to structure the project's analytical reasoning around incident distribution, asymmetry across segments, the relationship between severity and governance, and observable temporal evolution. In academic terms, this notebook provides the empirical evidence layer of the project.[1]

8.Machine Learning Modelling

Notebook 3 implements the Modeling phase of CRISP-DM and develops supervised models for severity classification and regulatory investigation prediction.[1]

8.1 Modelling Dataset

The notebook states that the modelling dataset contains 31 incidents, 19 initial features, and no missing values in the main columns. It also reports that the target distributions are extremely imbalanced, with severity_level concentrated in low severity cases and regulatory_investigation almost entirely concentrated in class 0.[1]

8.2 Selected Features

The modelling stage uses application_type, incident_type, customer_segment, year, regulatory_investigation, fine_imposed, policy_change, and third_party_audit as modelling variables. After one-hot encoding the categorical fields, the notebook records 14 input features for the models.[1]

8.3 Targets

The report creates a binary severity target where 0 corresponds to low or medium and 1 corresponds to high or critical severity, with distribution 28 versus 3. It also defines a multiclass target with four labels, but notes that the practical analysis relies mainly on the binary version because the multiclass setting is not viable under the extreme scarcity of examples.[1]

8.4 Implemented Models

The algorithms considered include Random Forest Classifier, XGBoost Classifier, and Logistic Regression as a baseline. This is presented as a conceptually appropriate comparison between a linear baseline and tree-based models, even though the major limitation remains the size and balance of the dataset.[1]

8.5 Train-Test Split and Imbalance

For the binary severity model, the notebook records X with dimension (31, 14) and y with dimension (31,), with 90.3 percent low severity and 9.7 percent high severity. The notebook explicitly identifies the dataset as imbalanced, with ratio 9.33, and recommends techniques such as class_weight, SMOTE, and balancing strategies.[1]

8.6 Reported Results

The final notebook summary reports XGBoost as the best severity classifier with F1-score 0.0000 and ROC-AUC 0.0833. For regulatory investigation prediction, it reports XGBoost with F1-score 0.0000 and ROC-AUC equal to nan.[1]

8.7 Correct Technical Interpretation

These results show that the modelling stage, in its current form, did not achieve reliable predictive performance. The technically correct reading is that the ML pipeline was implemented successfully, but the dataset is insufficient for robust predictive performance, so the work should be treated as a methodological proof of concept rather than a mature predictive solution.[1]

8.8 Analytical Value Despite Weak Performance

Even with weak results, Notebook 3 has academic and technical value because it demonstrates data preparation for ML, encoding and transformation of categorical

variables, target definition, algorithm comparison, metric-based evaluation, and serialisation for later API integration

9. Relational Database and RESTful API

Notebook 4 implements the Deployment phase by creating a Flask API to expose the project's data and models

9.1 API Objective

The API is designed to enable structured incident queries, access to aggregated statistics, integration with predictive models, and consumption by dashboards or external systems.[1]

9.2 Database Layer

The notebook uses a SQLite database with at least the main incidents table and complementary queries involving incidents, financial impacts, and regulatory responses. The notebook confirms 31 incidents in the database, consistent with the analytical base used in the other modules.[1]

9.3 Loaded Models

The deployment notebook attempts to load the artefacts severity_classifier.pkl, investigation_classifier.pkl, features_severity.pkl, and features_investigation.pkl. It

also records that the models expect 14 features, which is consistent with the Notebook 3 pipeline.[1]

9.4 Implemented Endpoints

The implemented endpoints include incident data retrieval, aggregated statistics by application, region, segment, governance, and time, as well as predictive endpoints for severity and investigation and a root endpoint for basic documentation.[1]

9.5 Architectural Role of the API

The API plays an important role because it converts analytical artefacts into a reusable service layer, moving the project closer to an applied architecture where notebooks are not only exploratory documents but also parts of a consumable pipeline.[1]

9.6 Critical Observation

Although the API infrastructure is well aligned with the project's goals, the practical value of the predictive endpoints depends on the quality of the serialised models. Since Notebook 3 reports F1 equal to 0 in the reported scenarios, the API should be interpreted as a demonstration of integration rather than a reliable decision mechanism.[1]

10. Interactive Dashboard

10.1 Objective

The dashboard is presented as the visual consumption layer of the project, allowing non-technical users to explore incidents, aggregated statistics, and predictive functionalities in an interactive way.[1]

10.2 Intended Integration

The intended architecture is sequential: Notebook 1 generates the treated base and database, Notebook 3 generates serialised models, Notebook 4 publishes the API, and the dashboard consumes the API endpoints.[1]

10.3 Suggested Pages

The source report suggests dashboard pages for incident overview, distribution by application type, customer segment analysis, governance and regulatory response, severity prediction, and investigation prediction.[1]

10.4 Recommended Improvements

For greater maturity, the report recommends authentication, pagination, robust error handling, API versioning, structured logging, caching, payload validation, and monitoring.[1]

11. System Architecture (Production-Grade MLOps)

11.1 General View

The architecture follows modern MLOps principles with a clear separation between data ingestion, preparation, modelling, and serving, while preserving a path for future evolution toward fully automated production pipelines.[1]

11.2 Architectural Components

11.3 Data and ML Flow

The flow described in the source moves from ingestion of incidents from AIID, Kaggle, and APIs into processing, feature engineering, training, model registration, storage, inference serving, and final dashboard consumption.[1]

11.4 Architectural Strengths

The architecture stands out for its separation between data, features, models, and serving, its readiness for integration with MLflow, Airflow, Docker, and Kubernetes, and its support for auditability, reproducibility, and AI governance.[1]

11.5 Evolution Roadmap

The roadmap proposed in the source includes adding MLflow for experiment tracking and model registry, automating the pipeline with Airflow, containerising services with Docker, deploying in cloud platforms such as AWS or GCP, and implementing model monitoring for drift, performance, and risk alerts.[1]

12. Technical Structure of the Notebooks

Notebook 1 is responsible for data acquisition and integration, cleaning, financial filtering, derived variable creation, and generation of the treated CSV and SQLite database. Notebook 2 is responsible for descriptive analysis, hypothesis testing H1 to H4, analytical visualisations, and interpretation of findings for risk and governance.[1]

Notebook 3 is responsible for feature preparation, encoding, target construction, model training and evaluation, and serialisation of artefacts for the API. Notebook 4 is responsible for SQLite connection, model loading, endpoint definition, consumption testing, and deployment structure.[1]

The report also mentions a possible Notebook 5 for pipeline consolidation, integrated documentation, or unified execution, but notes that this would be a future expansion if not yet implemented.[1]

13. Consolidated Results

13.1 What the Project Consistently Delivers

The project consistently delivers a thematic dataset of AI incidents in financial services, derived variables useful for risk and governance analysis, a statistical layer with clearly defined research hypotheses, a complete supervised modelling pipeline, serialisation and API exposure of artefacts, and a basis for dashboard visualisation.[1]

13.2 What Must Be Stated with Caution

The modelling results show that the pipeline was built correctly, but the analytical base is small and strongly imbalanced, the final reported performance is insufficient, and the models should not be presented as a robust production solution.[1]

13.3 Interpretive Synthesis

The greatest strength of the project lies in its end-to-end analytical architecture rather than in a highly accurate predictive model. Its central value is the demonstration of how a public and imperfect dataset can be transformed into an integrated ecosystem for preparation, analysis, modelling, persistence, service, and visualisation.[1]

14. Limitations and Methodological Considerations

14.1 Data Limitations

The report identifies a reduced base of only 31 incidents, strong class imbalance, dependence on publicised incidents, possible geographic and media bias, and heterogeneous textual granularity.[1]

14.2 Statistical Limitations

It also points to low statistical power for strong inference, high sensitivity to a few extreme cases, and fragile testing in small subgroups.[1]

14.3 Machine Learning Limitations

The ML limitations include classes that are too rare for robust generalisation, high risk of overfitting, and final metrics that are incompatible with serious operational use.[1]

14.4 Deployment Limitations

On the deployment side, the API is functional but lacks typical production controls, the predictive endpoints depend on fragile models, and the dashboard still requires consistency checks across endpoints and field names.

15. Execution Guide

15.1 Installation

The recommended installation command is pip install -r requirements.txt

15.2 Recommended Execution Order

The recommended order is notebook_1_exploracao_preparacao.ipynb, notebook_2_analise_estatistica.ipynb, notebook_3_modelagem_ml.ipynb, and notebook_4_api_restful.ipynb

15.3 Running the API

The report indicates running the API with python app.py, or with Gunicorn using gunicorn -w 4 -b 0.0.0.0:8000 app:app

15.4 Running the Dashboard

To run the interactive interface locally, the report instructs the user to go to the dashboard folder and execute streamlit run app.py, after which the application becomes available at http://localhost:8501

16. File Structure

The file structure described in the source includes dedicated folders for the API, dashboard, database, raw and processed data, models, notebooks, reports, and the root files requirements.txt and README.md

17. Dependencies

The main libraries listed in the source are pandas, numpy, scipy, statsmodels, scikit-learn, xgboost, joblib, flask, flask-cors, streamlit, plotly, matplotlib, seaborn, and sqlite3. The report also lists desirable dependencies for future expansion such as shap, pytest, pydantic, sqlalchemy, and redis

18. Conclusions and Next Steps

The project demonstrates a complete analytical architecture for the study of AI incidents in financial services, combining data preparation, statistical analysis, supervised modelling, relational persistence, RESTful API design, and interactive visualisation. Its main contribution lies in showing how a public and imperfect dataset can be transformed into a coherent technical pipeline for AI governance, compliance, and risk management support.

At the same time, the evidence shows that the available base is still limited for supporting robust predictive models. The ML components should therefore be positioned as proof of concept, while the analytical and architectural layers should be highlighted as the primary delivery of the project.

18.1 Expanded Conclusion

The consolidated reading of the project indicates that its most relevant contribution is not a highly accurate classifier, but the construction of an end-to-end analytical architecture capable of transforming public AI incidents into usable inputs for governance, risk, and compliance. Across the pipeline, the project connects data acquisition and treatment, derived variable construction, statistical analysis, supervised modelling, relational persistence, API exposure, and dashboard consumption, which gives the work an applied and technically coherent character.[1]

Substantively, the results support the idea that certain usage contexts are more exposed to failures with potential bias, operational damage, and governance fragility.

The statistical analysis suggests concentration of incidents in certain application types, as well as signs of unequal exposure across customer segments in cases associated with algorithmic bias

Another important point is the evidence of misalignment between potential severity and formal governance response in the observed base. Even when incidents show signs of higher severity, the available documentation indicates limited presence of regulatory investigation, policy change, independent audit, or other structured responses.[1]

Methodologically, the project also reveals the limits of working with publicised incidents and rare classes. The small base, strong imbalance, and dependence on heterogeneous textual fields reduce statistical power and compromise predictive robustness, so the correct interpretation is that the machine learning models function as a methodological proof of concept and architectural integration layer rather than mature operational instruments

This does not diminish the value of the work. On the contrary, it strengthens the academic quality of the project because it avoids inflated conclusions and presents an honest reading of empirical limitations

Under an applied lens, the project can be understood as a prototype for observability and analytical intelligence in algorithmic risk. In a real bank, fintech, insurer, or regulatory environment, an architecture of this type could support incident monitoring, audit prioritisation, policy review, impact analysis by customer segment, and executive communication about AI risk.

The strongest final conclusion is that the project delivers real value when positioned correctly: not as a final prediction product, but as an integrated analytical base for studying how AI failures emerge in financial services and how they relate to bias, operational damage, and governance

18.2 Recommended Next Steps

1. Increase the incident base and review labelling criteria.[1]

2. Refine derived variables with more robust NLP methods.[1]
3. Reassess class balancing with appropriate techniques.[1]
4. Strengthen validation of API endpoints and contracts.[1]
5. Add authentication, automated tests, and observability.[1]
6. Evolve the dashboard toward a more complete demonstration use case.[1]
1. Definitively standardise field names across notebooks, database, API, and frontend.[1]

19. References

- AI Incident Database (AIID)
- Official AIID GraphQL API
- Kaggle: AI Incident Database.
- Flask official documentation.
- Streamlit official documentation.
- Scikit-learn documentation.
- XGBoost documentation
- CRISP-DM methodology reference
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